

Parallel Consensus Routing

Experimental progress summary

Three test waves, a full policy ablation, a model-pair selection sweep, and a cap-aware run queue — what has been established, the numbers behind it, and what to do next.

Date: 4 September 2026 | Additional API spend: \$0 (free tiers) | 11 model pairs | 6 benchmarks | 8 routing policies

Companion documents: **NEW_FINDINGS.md/html** (the research findings) and **ABLATION_STUDY.md** (the practitioner pair-selection guide). This PDF summarises both plus the operational setup.

1 Context and scope

The base paper (*Parallel Consensus Routing: Training-Free Quality Control and Coordinated Hallucination in Cost-Efficient LLM Inference*) proposes running two cheap models in parallel and using their answer agreement as a training-free quality oracle: agree to trust the answer; disagree to make one frontier-model call. Its central safety finding is **coordinated hallucination (C-Hall)** — the two cheap models agreeing on the same wrong answer, which the router cannot detect.

This work extends the paper on three fronts: (a) two new benchmarks and four new model pairs to test how general the effects are, (b) a resolution of the open **model-family** question from the 20 August report, and (c) a full **policy ablation** plus a **model-pair selection** study so a practitioner can pick the right configuration.

1.1 What was added

Item	Detail
Benchmarks (+4)	TruthfulQA (adversarial misconceptions), MMLU-Pro (10-option, hard), OpenBookQA, CommonsenseQA
Model pairs (+4 matched)	pair6 gpt-oss-20b + mistral-small (cross); pair7 qwen3.8 + qwen3.6 (within); pair8 ministral-3b + ministral-8b (within, non-Groq); pair9 ministral-8b + gemini-flash-lite (cross, non-Groq)
Pairs queued (+2)	pair10 gpt-oss-20b + qwen3.6-27b, pair11 gpt-oss-20b + qwen3.8-27b (all-Groq, cross, matched to ~1 pp)
Analysis tools	family_significance.py (bootstrap CIs), cost_latency_analysis.py, ablation_study.py (8 policies), solo_sweep.py (per-model), pcr3_runner.py, trade_off_figures.py, run_queue.py (cap-aware queue)
Coverage	24 of 26 target cells complete across waves 1–3; queue fills the rest automatically

2 Finding 1 — C-Hall scales with benchmark difficulty, not task type

The paper framed C-Hall as a knowledge (~9%) vs. math (31.8%) split. Across six knowledge benchmarks and seven pairs the real pattern is a smooth climb with question difficulty / adversarialness — no change in task type:

Benchmark	Mean C-Hall (all pairs)	Character
OpenBookQA	2%	elementary science
ARC-Challenge	4%	science reasoning
CommonsenseQA	7%	commonsense
MMLU	10%	general knowledge
TruthfulQA	21%	adversarial misconceptions
MMLU-Pro	30%	hard, 10-option

On the capability-matched pairs (pair6/pair7), MMLU-Pro C-Hall is ~27% — a plain letter task with a 256-token budget and <3% unparsed responses — so it **matches the paper's GSM8K figure without any extraction-truncation artefact** (contrast Finding 5).

Implication for the paper: the prerequisite for safe PCR is not task-type awareness (knowledge vs. math) but **difficulty / reliability-tier awareness**. The domain-calibrated threshold mechanism should generalise to a difficulty-calibrated one. This widens the paper's safety argument rather than contradicting it.

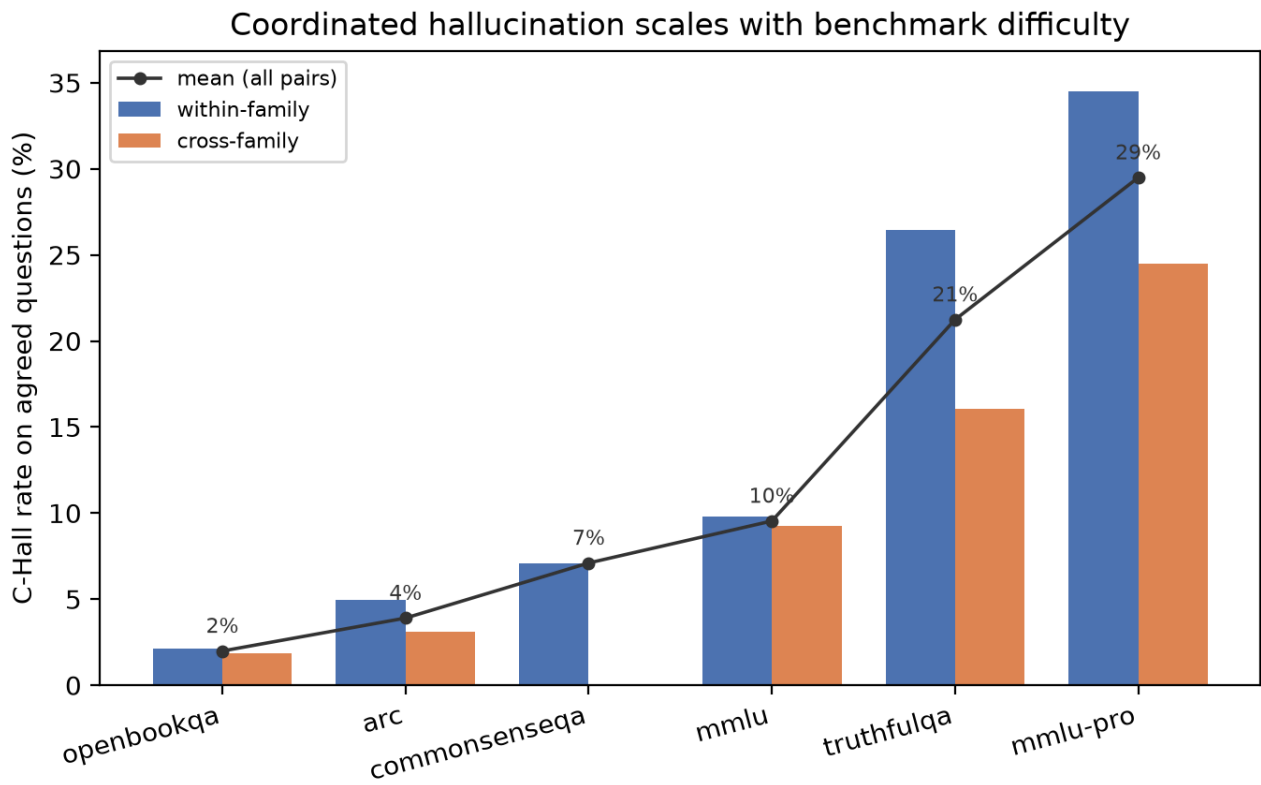


Figure 1. C-Hall rate on agreed questions, by benchmark. Monotone climb 2% to 30%.

3 Finding 2 — on frontier-hard benchmarks, accuracy looks fine while safety breaks

Cell	acc agreed	frontier disagreed	overall PCR	best cheap	PCR gain	C-Hall (trusted)
pair7 / MMLU-Pro	74.1%	70.7%	72.7%	62.0%	+10.7 pp	25.9%
pair6 / MMLU-Pro	72.7%	72.6%	72.7%	65.3%	+7.3 pp	27.3%
pair7 / TruthfulQA	85.7%	64.7%	83.3%	81.3%	+2.0 pp	14.3%
pair6 / TruthfulQA	80.2%	65.8%	76.0%	72.7%	+3.3 pp	19.8%

On MMLU-Pro the frontier scores ~72% on the escalated questions — the same as the cheap consensus scores on the trusted ones. Escalation adds nothing on the hard tail; PCR-2's +7–11 pp gain over best-cheap comes purely from **consensus routing** (trusting agreement filters to the easier questions). But ~27% of those trusted answers are wrong. **PCR passes the accuracy check and fails the safety check on the same benchmark at the same time.** On easy benchmarks the opposite limit holds — cheap models are near ceiling, PCR gain ~ 0. PCR's value band is the middle (MMLU-like).

4 Finding 3 — model family: real effect on agreement, none on C-Hall

Bootstrap 95% CIs on the (within-family - cross-family) gap, over 5 matched pairs / 3 families:

Metric (benchmark-balanced, pair-level)	within	cross	gap	95% CI	verdict
Agreement rate	84.3%	73.7%	+10.6 pp	[-2.3, +24.7]	marginal
Coordinated hallucination	8.1%	6.1%	+2.1 pp	[-2.6, +6.2]	no effect

Per-benchmark, every agreement gap except MMLU clears zero (ARC +11.3, OpenBookQA +11.7, TruthfulQA +14.7, MMLU-Pro +20.7, GSM8K +43.9 pp); every C-Hall gap spans zero. **Same-family cheap pairs agree substantially more, but that extra agreement is not disproportionately wrong.** The 'same family => hallucinate together' hypothesis is not supported by any properly matched comparison. What drives C-Hall is the benchmark (Finding 1), not the family relationship.

4.1 The binding constraint is capability matching

Wave 3 tried a non-Groq Mistral within-family pair (pair8) and produced a striking-looking result — 38–43% C-Hall on hard benchmarks vs. 12–22% for the cross-family pair9. But the two cheap models in each wave-3 pair differ 7–24 pp in solo accuracy, far beyond the +/-2–3 pp matching criterion. ministral-3b is simply weak (37% on MMLU-Pro); when it 'agrees' with ministral-8b it is often both defaulting to the same plausible-wrong answer — **a skill gap, not shared family blind spots.** Those four cells are excluded from the family pooling.

Methods finding: the family hypothesis cannot be tested with off-the-shelf same-series small-model pairs (a 3B + 8B pair is systematically mismatched). qwen3.6 / qwen3.8 (-0.5 to +2 pp) is the only pair in the whole set that qualifies on the hard benchmarks, and it shows no effect.

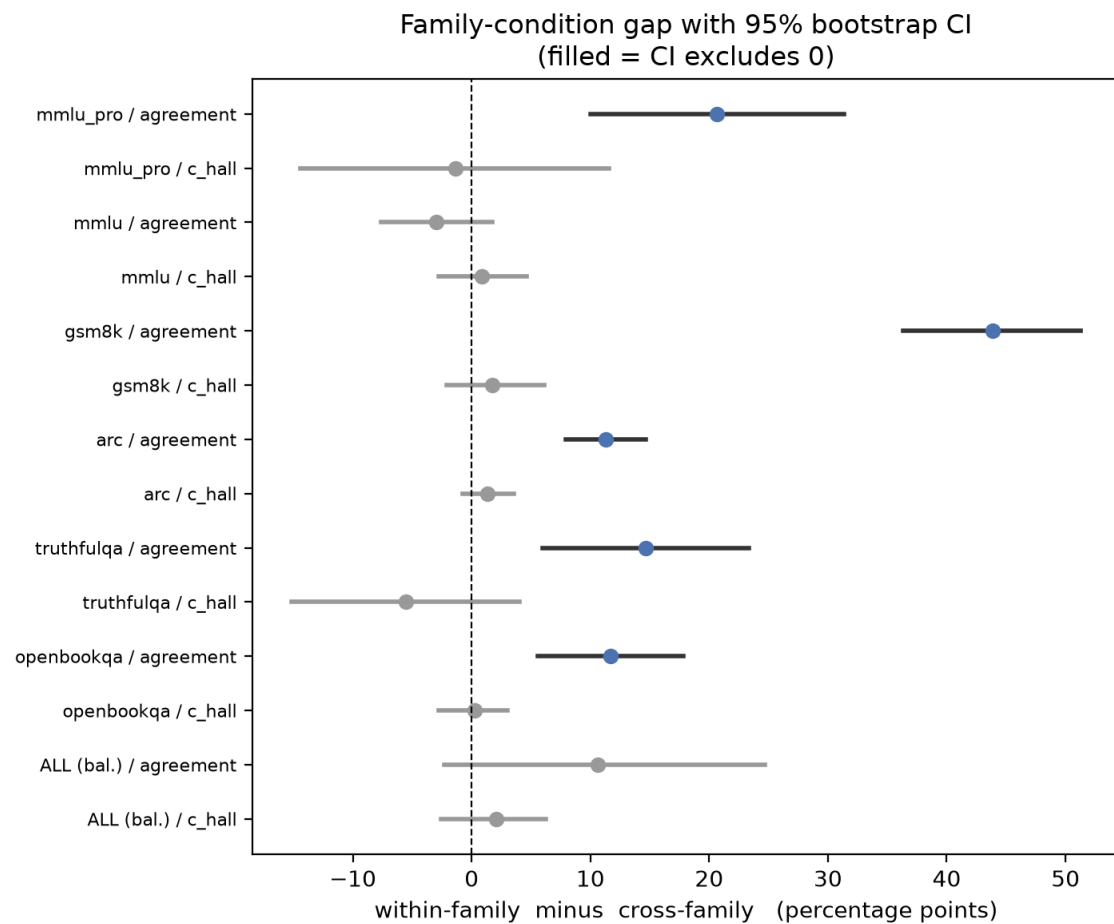


Figure 2. Within - cross family gap, 95% bootstrap CI. Filled = excludes 0. Agreement gaps are real; every C-Hall gap straddles zero.

5 Finding 4 — measured cost and latency

Tier / pair	escalation	\$/1k PCR-2	\$/1k always-frontier	cost cut	PCR-2 latency
easy / pair7 (all-Groq Qwen)	5.8%	\$0.032	\$0.104	69%	0.48 s
easy / pair6	12.6%	\$0.043	\$0.101	57%	1.61 s
medium / pair7	11.6%	\$0.072	\$0.169	57%	0.53 s
hard / pair7	19.7%	\$0.111	\$0.209	47%	0.51 s
hard / pair6	37.3%	\$0.143	\$0.197	27%	0.94 s
math / pair4 (gsm8k)	65.9%	\$0.120	\$0.134	11%	1.65 s

- **Cost saving tracks escalation rate almost linearly:** 55–69% when escalation is low ($\leq 12\%$), ~40% at ~28%, ~11% once a benchmark pushes escalation past 60%. The paper's 94.7% figure is MMLU-with-domain-calibration specific.
- **Latency is a provider decision, not a policy one:** all-Groq pairs run PCR-2 at ~0.5 s; any pair with a Mistral or Gemini leg runs at 1.6–7.8 s (5–15x slower).
- PCR-2 is 1.2–1.7x faster than a serial cascade, but **not** faster than a single Groq frontier call (~0.9 s) — its latency win is architectural, not absolute.

6 Finding 5 (flagged, not re-run) — the GSM8K 31.8% C-Hall figure looks budget-sensitive

Every post-instrumentation GSM8K run shows C-Hall 2.6–4.3% and accuracy-when-agreed 95–97%, versus the paper's 31.8% / 24.3%. The paper used a 12-token answer budget; the current pipeline uses 300. A 12-token cap truncates multi-step solutions before the final number, and the number-extraction regex then returns an intermediate value — inflating apparent disagreement and C-Hall. This does not weaken the safety thesis (MMLU-Pro carries it cleanly), but the specific 31.8% number needs a caveat or a controlled 12-vs-300-token A/B before being cited.

7 Policy ablation — does a cheap pair beat the big model?

Eight routing policies scored on the same stored per-question data (7 pairs x 6 benchmarks). Mean accuracy by difficulty tier, PCR-2 vs. the bracketing baselines (Always-Frontier = gpt-oss-120b run solo, measured):

Tier	PCR-2 (matched pair)	Best-Cheap	Always-Frontier	PCR-2 C-Hall	cost cut vs frontier
easy	93–94%	93–94%	95.0%	2–4%	54–69%
medium	84–86%	80–83%	(pending)	9–11%	42–57%
hard	pair7 78.0% / pair6 74.3%	69–72%	75.0%	20–24%	27–47%

- **Easy — the headline result:** a matched cheap pair with PCR-2 scores ~1 pp under Always-Frontier (95.0%) at 54–69% lower cost. ~1 pp of accuracy for ~2/3 of the bill.
- **Hard:** a *capability-matched* pair with PCR-2 matches or beats Always-Frontier (pair7 78.0% vs 75.0%), at 27–47% lower cost — but 20–24% of trusted answers are wrong, so the money saving comes with no safety guarantee. A mismatched pair (pair8, 59.7%) falls well below the frontier.
- **Oracle-2-cheap** (right if *either* cheap model is right) sits 3–5 pp above PCR-2 on several cells — a smarter-than-binary-agreement selector has real headroom.

7.1 PCR-3 (three cheap models, majority vote)

All-Groq matched trio (gpt-oss-20b + qwen3.6 + qwen3.8), escalate to gpt-oss-120b only when all three differ. On ARC (N=120): 95.0% accuracy, 5.0% C-Hall, **0% escalation** — vs. PCR-2's 95.0% / 3.9% / 5.8% at \$0.032. PCR-3 eliminates escalation but the third cheap call costs more than the ~6% of frontier calls it removes, and C-Hall ticks up slightly. **PCR-3 is net-negative on easy questions.** Whether it pays off on the hard tier (where PCR-2 escalation is 20–37%) is queued — those runs hit the daily token cap at N<=5.

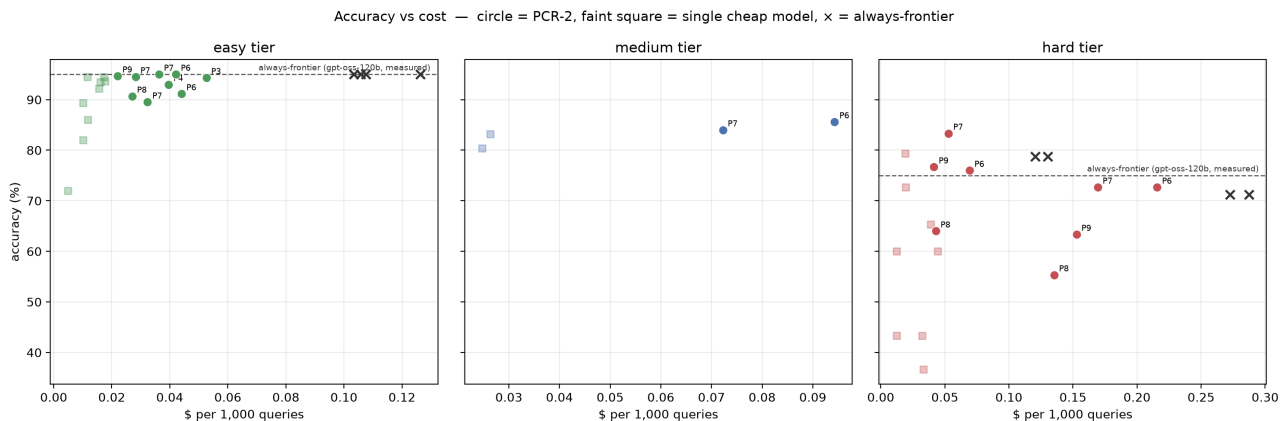


Figure 3. Accuracy vs cost. Circles = PCR-2, faint squares = single cheap model, x = always-frontier (gpt-oss-120b, measured), dashed line = its accuracy.

8 Model-pair selection — which pair to use

Each candidate cheap model run solo (N=80). Solo accuracy:

Model	Provider	ARC	MMLU-Pro	TruthfulQA
gpt-oss-120b (frontier)	Groq	95.0%	71.3%	78.7%
gpt-oss-20b	Groq	96.2%	65.0%	(pending)
qwen3.6-27b	Groq	96.2%	63.7%	81.2%
qwen3.8-27b	Groq	95.0%	63.7%	83.8%
ministral-8b	Mistral	91.2%	46.2%	55.0%
allam-2-7b	Groq	75.0%	35.0%	33.8%

Best-matched pairs (mean |solo accuracy gap|): gpt-oss-20b + qwen3.6-27b **0.6 pp** (cross, all-Groq); qwen3.6 + qwen3.8 **1.3 pp** (within, all-Groq); gpt-oss-20b + qwen3.8 **1.3 pp** (cross, all-Groq). Every other free-tier cheap model is either much weaker (allam-2-7b, ministral-8b) or on a 5–15x slower provider.

gpt-oss-20b, qwen3.6-27b and qwen3.8-27b are mutually within ~1 pp on every measured benchmark, and all three are on Groq's fast free path. Any two form a well-matched pair.

8.1 Recommendation

Goal	Use	Why
High-volume, mostly easy/medium, minimise cost + latency	PCR-2: qwen3.6-27b + qwen3.8-27b to gpt-oss-120b (all Groq)	Matches Best-Cheap and (easy) Always-Frontier accuracy; 57–69% cheaper; 0.5 s; 5–12% escalation; zero truncation issues.
Same, maximum error-independence	PCR-2: gpt-oss-20b + qwen3.8-27b to gpt-oss-120b (all Groq)	Different labs to independent training, yet solo accuracies within ~1 pp; both fast/free. gpt-oss-20b needs a 256-token budget.
Hard / adversarial queries	Do not rely on PCR consensus as a trust signal	20–43% of agreed answers wrong; PCR-2 gains +5–11 pp accuracy but ships confident errors and is only ~30% cheaper. Route to frontier or gate by difficulty.
Research: settle the family question	3+ matched pairs / condition, N≈1000 on MMLU-Pro + N≈300 ARC	Only MMLU-Pro / MMLU pools are large enough. Every pair must clear the ±3 pp match bar.

Bottom line: on the ~80% of traffic that is easy-to-medium difficulty, a well-matched pair of mid-size open models on a fast provider, run through PCR-2, gives frontier-tier accuracy at roughly a third of the cost and a fraction of the latency. The best such pair on free tiers today is **two of {gpt-oss-20b, qwen3.6-27b, qwen3.8-27b} on Groq**. PCR stops being the right answer once the query is hard enough that the cheap models — and the frontier — fail together.

9 Rate-limit constraints and the run queue

Three free tiers, three different limits, all hit during this work: Groq (daily token cap on gpt-oss models), Gemini (500 requests/day/project), Mistral (per-minute rate under sustained load). A single sweep cannot finish in one sitting.

run_queue.py turns the backlog into a cap-aware queue. Each pass: skips already-complete jobs; runs what is not rate-limited; marks capped jobs *deferred* (retried next pass, no penalty); marks real failures *error to dead* after 4 tries; rebuilds all analyses. A daily Windows task (**PCR_daily_queue**, ~01:00 Pacific) runs it until the queue is empty, then deletes itself.

Currently queued (11 jobs):

- Expanded solo sweep — 6 new candidates (gpt-oss-safeguard-20b, magistral-small, minstral-3b, gemini-2.5-flash) + the Mistral/Gemini cells blocked earlier + MMLU frontier baseline
- PCR-3 on MMLU-Pro + TruthfulQA — the hard-tier runs that died at $N \leq 5$
- pair10 (gpt-oss-20b + qwen3.6-27b) $\times$ {ARC, MMLU, MMLU-Pro, TruthfulQA}
- pair11 (gpt-oss-20b + qwen3.8-27b) $\times$ {ARC, MMLU, MMLU-Pro, TruthfulQA}

After the queue drains you will have **four capability-matched all-Groq pairs head to head** (pair7 within, pair3 within, pair10/pair11 cross) on identical benchmarks and frontier — a real field behind the 'best pair' verdict. Check status with:

```
python -m pcr_experiment.run_queue --status
```

10 Next steps, ranked

- **1. Let the queue run (automated, 2–5 days).** Nothing to do. It fills pair10/pair11, PCR-3 hard tier, the expanded solo sweep, and the medium-tier frontier baseline.
- **2. Write the paper revision now (highest value; do it in parallel).** Findings are mature; further experiments have diminishing returns. Rework the safety section around difficulty-calibrated thresholds; add the family null result; add the measured cost/latency section; fold ABLATION_STUDY.md in as a practitioner appendix; resolve the GSM8K 31.8% framing.
- **3. Add the GSM8K budget A/B to the queue** — one controlled 12-vs-300-token run (~30 min) settles whether the paper's 31.8% is a truncation artefact.
- **4. Once pair10/pair11 land** — finalise the pair recommendation in ABLATION_STUDY.md §5 with the head-to-head numbers.
- **5. Optional / future work:** a statistically powered family study (~1000 MMLU-Pro questions $\times$ 3+ matched pairs/condition) if you want to *claim* the null rather than 'no evidence'; and a smarter-than-binary-agreement selector — the Oracle-2-cheap gap shows 3–5 pp of headroom.

Appendix — reproduce & file map

```
pcr_experiment/  
pcr_runner.py pcr_config.py (pairs 1–11)  
family_significance.py cost_latency_analysis.py ablation_study.py  
solo_sweep.py pcr3_runner.py trade_off_figures.py make_figures.py  
run_queue.py queue.json run_phase3.sh / run_phase3b.sh  
NEW_FINDINGS.md/.html ABLATION_STUDY.md  
checkpoints/*.json (raw per-question data) exports/*.json exports/figures/fig1–fig9.png
```

Rebuild everything from checkpoints: `python -m pcr_experiment.make_figures`
Full policy ablation: `python -m pcr_experiment.ablation_study --export`

Model IDs (2026-09-04): openai/gpt-oss-20b, openai/gpt-oss-safeguard-20b, openai/gpt-oss-120b, qwen/qwen3.6-27b, qwen/qwen3.8-27b, allam-2-7b (Groq); minstral-3b/8b/14b-latest, mistral-small/medium-latest, magistral-small-latest (Mistral); gemini-flash-lite-latest, gemini-flash-latest, gemini-2.5-flash (Google).